ARCHIT AGRAWAL

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PROFESSIONAL EXPERIENCE

EdPlus, Arizona State University **06/2026 - Present**
AI Software Engineer *Phoenix, AZ*

- Validated 25 statistical tests and six causal estimators against SciPy and statsmodels; scored 68% on a 100-question StatQA sample, against GPT-4o's best reported 64.83%.
- Designed a controlled experiment showing model choice moves text-classification F1 33x more than pipeline design (0.230 vs 0.007), redirecting the team's roadmap toward reliability.
- Quadrupled the headline findings automated reports recover, 13% to 52–59% on held-out datasets, by letting an LLM choose analyses while code computes every number.
- Designed conformal accept and abstain thresholds that bound the error rate on accepted labels, giving every label a calibrated 0–100 confidence.
- Designed blind random QC audits per run that yield both a quality score and a corrected estimate published with its margin of error.
- Retired a 0.637 baseline the project's claims rested on after fresh runs reproduced it at 0.523, tracing the gap to a human-reviewed delivery file.
- Cut cross-run disagreements from 30,405 to 263 across 411 runs by giving every column one meaning in one semantic model.
- Calibrated codebook fit before labeling: a 25-row sample separated the right codebook at 0.598 with 4% abstention vs 0.316 and 88%, for about $0.001.
- Improved accuracy from 60% to 93% on the hardest benchmark questions by consolidating column definitions in a shared semantic layer.
- Designed a chart recommender covering 17 chart types, retaining deterministic rules when LLM proposals failed to improve evaluation scores.
- Blocked double counting in cross-dataset joins by detecting runs that share the same people.

MyStage Music Inc. **03/2025 - 06/2026**
Founding AI/ML Engineer *Remote*

- Lifted dataset accuracy 25% with Gemini entity resolution, scored by LLM-as-judge, over 70,000+ records a day.
- Scaled ingestion to 70,000+ records daily from 650+ venues using Playwright, transactional locking and entity resolution.

EdPlus, Arizona State University **09/2023 - 05/2025**
Software Engineer *Phoenix, AZ*

- Gated every prompt release on Prompt Flow LLM-as-judge evals for groundedness, relevance and coherence.
- Compared fixed-size and section-aware chunking on course documents to preserve complete policy passages in retrieved results.
- Cut transcript-analysis time from four hours to 15 minutes with a Neo4j knowledge graph and schema-validated, LLM-generated Cypher queries.

Arizona State University **08/2024 - 05/2025**
Student Researcher *Phoenix, AZ*

- Built a PyTorch and Diffusers benchmark for text-to-video models that scores physical plausibility separately from visual quality, holding seed, resolution and sampler fixed.
- Found visual quality barely predicts physical plausibility: the top model by human preference ranked last on causal consistency.
- Designed a frame-level rubric for rigid-body motion, gravity, momentum and occlusion, and measured inter-rater agreement before trusting any aggregate score.
- Derived a failure taxonomy from annotated clips: broken object permanence, non-conserved mass, teleporting contact and physics that resets at a scene cut.

Knowledge Exchange for Resilience, Arizona State University **06/2024 - 08/2024**
Data Research Aide *Phoenix, AZ*

- Built a FastAPI and pgvector collaborator recommender achieving 85% top-k precision in evaluation.
- Lifted search relevance 15% over baseline with an n-gram ranking model over cleaned researcher-profile text.
- Reduced p95 retrieval latency 60% at 95% recall using hybrid FAISS and Neo4j search.

Zeus Learning **01/2022 - 07/2023**
Software Engineer *Mumbai, India*

- Built a desk-demand forecasting model on historical occupancy by floor, day and team, served from Redis to 300+ offices; reported occupancy improved 30%.

DA-IICT **05/2021 - 08/2021**
Computer Vision Researcher *Gandhinagar, India*

- Engineered an autonomous-driving perception stack in Python: radar and lidar capture, timestamp alignment, extrinsic calibration and point-cloud processing, with a replay harness over recorded drives.
- Compared early and late sensor fusion on identical sequences, showing point-level merging keeps detail but inherits both sensors' noise while object-level merging is robust but discards evidence.

PROJECTS

Soccer Result Prediction **10/2023 - 12/2023**

- Lifted match-outcome accuracy 12% by ensembling LSTM and XGBoost over rolling-window form features, with a time-based split that prevents leakage.
- Scored models on log loss, checked calibration with reliability plots and reported per-class recall for the hard draw class.

Jet Engine Remaining Useful Life **10/2024 - 12/2024**

- Predicted jet-engine remaining useful life from NASA CMAPSS sensor time series, engineering features and comparing machine learning, deep learning and generative regressors.

AiJockey – AI DJ Pipeline | GitHub **2025 - Present**

- Built a PyTorch/FastAPI audio pipeline with stem separation, LLM transition planning and 25+ DSP transitions; ported inference to AMD MI300X.
- Trained a MERT-95M reward head to score audio segments across four aesthetic dimensions for candidate selection.
- Fine-tuned with DPO, KTO and IPO preference training; reduced DPO training loss from 0.68 to 0.47 on labeled pairs.

TECHNICAL SKILLS

Languages & Libraries: Python, SQL, pandas, NumPy, scikit-learn, XGBoost, SciPy, statsmodels, PyTorch, TensorFlow, Diffusers

Statistics: Hypothesis testing, A/B testing, experiment design, causal inference, regression, survey weighting, sampling, margin of error, conformal prediction

Machine Learning: Classification, NLP, text classification, embeddings, clustering, recommendation, ranking, time series, forecasting, deep learning (LSTM, CNN), fine-tuning (DPO)

Evaluation: Model evaluation, benchmark design, log loss, F1, precision@k, recall, calibration, inter-rater agreement, LLM-as-judge

Data & Visualization: DuckDB, PostgreSQL, pgvector, FAISS, Neo4j, AWS Bedrock, Plotly, D3, Vega-Lite

EDUCATION

Arizona State University (ASU) **08/2023 - 05/2025**
M.S., Computer Science | GPA: 4.0 *Tempe, AZ*

Dhirubhai Ambani Institute of Information and Communication Technology **08/2018 - 05/2022**
B.Tech., Information and Communication Technology *Gandhinagar, India*